

# The Beneficent and Maleficent Effects of Simplification on Retirement Savings

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Considerable research suggests that making information simpler is better. Simplification improves the efficiency of information extraction and lowers psychological frictions, leading to its popularity with policymakers and practitioners worldwide. However, it remains unclear when and how simplification can be utilized most effectively, or if there are contexts where simplification may produce unintended maleficent effects. Using two large-scale field experiments ( $N = 126,673$ ), we test whether simplifying account statements helps encourage retirement savings in Mexico. We partner with two retirement firms, one ranked high in rate of returns and the other ranked lower. We find that simplifying retirement account statements improves contribution rates for consumers in the high-ranking firm but reduces contribution rates for consumers in the low-ranking firm. Five follow-up experiments provide evidence consistent with a fluency amplification account. Simplifying information improves processing fluency, making it easier to accurately recall firm rank relative to the control, which amplifies behavior bidirectionally: High-ranking (low-ranking) firm consumers more accurately recall their firm's rank, subsequently increasing (decreasing) contributions. However, if simplification is harnessed in ways that improve processing fluency and lower perceived switching costs, then simplification can improve retirement savings for everyone either by boosting contributions or encouraging people to switch to higher performing alternatives.

**Keywords:** cognitive complexity, simplification, financial decision-making, field experiment, heterogeneous treatment effects

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*"Everything should be made as simple as possible, but not simpler."* —Albert Einstein

Many parts of the world suffer from savings gaps, where citizens do not save at the rate needed to produce well-funded retirements. Small barriers and obstacles, such as complexity in understanding the benefits of savings or not knowing how to make a retirement contribution, can prevent individuals from securing a better financial pathway for the future. Interventions designed to simplify the way information is presented are one solution that has been applied to other decision contexts where individuals are facing similar hurdles (Sunstein 2013; Thaler and Sunstein 2008).

The use of simplification for redesigning basic forms, notices, or account statements—for example, using plain language, sorting alternatives on a particularly salient attribute, including visual aids and graphics to display information, and/or providing clear actionable steps—has become increasingly popular with governments and nudge units globally. In fact, a recent meta-analysis reviewing various behavioral interventions found that simplification is the single most popular intervention used by governments and nudge units, applied in 36% of randomized controlled trials (DellaVigna and Linos 2022).

Governments and organizations send out forms and statements that can be overly complex and difficult to process. If simplification can help to improve information comprehension while also lowering the amount of cognitive effort required, then simplifying existing communications provides a low-cost, easy-to-implement solution with little, if any, harmful consequences (Sunstein 2013; Thaler and Sunstein 2008). There is also abundant evidence that simplification can improve a variety of behaviors from increasing the take-up of social assistance and vaccination rates, to improving school choice and financial aid application submissions (Bettinger et al. 2012; Bhargava and Manoli 2015; Glazerman 2017; Milkman et al., 2011).

Here, we apply simplification to the domain of retirement savings. We ask whether simplifying retirement account statements encourages individuals to save more for retirement. In experiments 1A and 1B, we partner with two Mexican retirement firms, running two parallel, large-scale field experiments: One firm ranked high in expected returns relative to other firms, while the other ranked in the bottom third. We hypothesized that simplifying the retirement account statements would increase the likelihood of voluntary contributions. Our hypothesis held for customers in high-ranking firms. However, simplifying account statements significantly decreased contributions for low-ranking firm customers.

The heterogeneity in response to our simplification interventions was not anticipated by the research team or in post-test surveys with policymakers, marketing communication and design experts, and a general population sample from Mexico. In a follow-up survey with policymakers and

marketing experts ( $N = 74$ ), 70.3% predicted that the simplified form would improve contributions regardless of the firm and 29.7% predicted that redesigned forms would make no difference to contribution due to inattention. Zero predicted a backfire effect (web appendix A, study 1A). A different sample of participants who had policy, government, marketing, and education design experience ( $N = 391$ ) also did not predict a backfire. Shifting from a more complex form to a simplified form was predicted to substantially increase voluntary contributions for both low-ranked and high-ranked firms (web appendix A, study 1B). It is possible that those working in policy and marketing may be less accurate in predicting end-user behavior relative to a general sample that typically receives those specific forms.

To address this possibility, we ran a prediction survey with a general population sample from Mexico ( $N = 200$ ). We asked Mexican citizens to predict which of the two forms—the standard statement or simplified, redesigned statement—would have higher voluntary contribution rates, varying whether people were sent a statement from a firm ranked first in terms of rate of return or one ranked last. Once again, while the beneficial effects of simplification were predicted when the firm was ranked first (76% predicted that the simplified form would yield higher contributions), the maleficent effects were not, with 71% predicting that the simplified form would yield higher contributions despite the firm being ranked at the bottom (web appendix A, study 1C).

We use the results from our two field experiments to develop a framework to understand why and how simplification might produce the heterogeneous treatment effects we observe. Consistent with research on processing fluency and fluency amplification (Albrecht and Carbon 2014; Landwehr and Eckmann 2020), we hypothesize that simplification improves the ease of processing information and efficiency of information extraction, but that fluency can have an amplifying effect on evaluation and behavior. We predict that when simplification improves the recall of positive information (e.g., the consumer's firm is ranked highly), then voluntary savings contributions will increase. Alternatively, if simplification improves the recall of negative information (e.g., the firm is ranked low), then voluntary savings contributions will decrease. We conduct several robustness checks using follow-up laboratory experiments to validate our assumptions, rule out potential alternative explanations, and replicate the heterogeneous treatment effects observed in the field. We also test for the potential process underlying our effect and find that simplification may unintentionally discourage contributions for those in low-ranked firms by making negative information easier to process and recall.

In addition to increasing voluntary savings, it is possible to improve retirement savings via another route: switching to a firm with a higher rate of return. However, if consumers are unaware that they can switch firms or they perceive

the cost of switching to be too high, then interventions focused on making the account statement information easier to process will not motivate those in lower-ranked firms to switch to higher performing options. But if simplification is applied in ways that improve information acquisition and lower perceived switching costs, for example, simplifying the information of how to switch to a better performing option, then simplification could be an efficient and low-cost tool for increasing the savings amount from mandatory contributions. We test this possibility in a follow-up laboratory experiment. Table 1 provides an overview of our experiments, participants, and findings.

Our findings make four main contributions. First, we demonstrate how and when simplification and processing fluency produce amplification effects and shape consumer behavior at scale. Second, we demonstrate how and when simplification and processing fluency produce amplification effects. Using evidence from a follow-up process-tracking experiment, we show that simplification operates by improving the efficiency of information processing. Consumers recall their firm rank more accurately despite spending the same time or even less time on the given information. However, we find that improving the processing fluency has an amplifying effect depending on the valence of information.

Third, from a substantive perspective, our field experiments and follow-up lab experiments suggest that simplifying and improving the ease of processing information can influence consequential behavioral outcomes, such as saving for retirement. Most research investigating fluency effects has focused predominantly on evaluation valence and judgments of liking. Less is known about whether improving

fluency and the ease of processing can extend to behavioral outcomes as well. We demonstrate how these effects operate at scale and, in a setting, where saving for retirement is not the norm—with typically less than 0.5% make a voluntary savings contribution annually (Fertig et al. 2015).

Fourth, our insights have clear implications for policy. We use our results to develop a framework for how policy-makers and practitioners can effectively utilize simplification to improve consumer welfare and well-being. When simplification is applied in ways that make the multiple pathways to improve retirement savings salient, then simplification can be an effective tool that can boost savings overall. Finally, our work contributes to the growing literature leveraging psychological insights in ways that can inform existing public policy (John et al. 2017; Johnson and Goldstein 2003; Keller et al. 2011; Sharpe, Staelin, and Huber 2008). Next, we review research on how fluency affects evaluations and information processing bidirectionally to help build our predictions.

## CONCEPTUAL BACKGROUND

### Fluency Affects Evaluations and Information Processing

Extant evidence finds that individuals have limited cognitive and processing capabilities (Payne et al. 1993; Simon 1955, 1956). Processing fluency, or the subjective experience of ease with which people process information, can affect perceptions and memory processes in ways that can help people assess choices and make decisions efficiently (Simon 1990; Tversky and Kahneman 1974). Making stimuli easier to process can also influence human judgment by increasing

TABLE 1  
OVERVIEW OF FINDINGS ACROSS ALL STUDIES

| Study                                                        | Participant sample                            | Result                                                                                                                                                                                                           |
|--------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Study S1A, appendix A<br><i>Prediction survey</i>            | Policy and marketing experts<br><i>N</i> = 74 | 70.3% predict beneficent effects of simplification<br>0.0% predict backfire effects                                                                                                                              |
| Study S1B, appendix A<br><i>Prediction survey</i>            | Prolific online sample<br><i>N</i> = 391      | Shifting from complex to simplified form predicted to boost savings<br>Predicted increase for both high-rank and low-rank firms                                                                                  |
| Study S1C, appendix A<br><i>Prediction survey</i>            | Mexico general sample<br><i>N</i> = 200       | 76% predict simplification will improve contributions (high rank)<br>71% predict simplification will improve contributions (low rank)                                                                            |
| Field experiment 1A:<br><i>High-rank firm</i>                | Mexican citizens<br><i>N</i> = 62,796         | Simplified forms increase contribution likelihood<br>Simplified forms increase amount contributed                                                                                                                |
| Field experiment 1B:<br><i>Low-rank firm</i>                 | Mexican citizens<br><i>N</i> = 64,870         | Simplified forms decrease contribution likelihood<br>Simplified forms decrease amount contributed                                                                                                                |
| Experiment 2<br><i>Process evidence</i>                      | Prolific online sample<br><i>N</i> = 725      | Rank simplification increases rank recall accuracy<br><i>High rank</i> : Rank simplification increases contributions<br><i>Low rank</i> : Rank simplification decreases contributions                            |
| Study S2, appendix E<br><i>Assumption validity</i>           | Amazon Mturk<br><i>N</i> = 302                | Simplified forms significantly ease processing fluency<br>Simplified forms increase rank recall accuracy                                                                                                         |
| Study S3, appendix F<br><i>Shift in attribute importance</i> | Amazon Mturk<br><i>N</i> = 401                | Rank simplification increases importance of firm ranking<br>Customer service and operational effectiveness less important                                                                                        |
| Study S4, appendix G<br><i>Simplifying firm switching</i>    | Prolific online sample<br><i>N</i> = 565      | When both rank and switching information is simplified:<br>Low rank consumers more likely to switch to better firm                                                                                               |
| Study S5, appendix I<br><i>Robustness check</i>              | Amazon Mturk<br><i>N</i> = 303                | Information valence causes amplification effects:<br><i>Low rank versus no rank info</i> : Rank info decreases contributions<br><i>High rank versus low rank and no rank</i> : Rank info increases contributions |

the likelihood that stimuli are evaluated more favorably (Bornstein 1989; Jacoby 1983). For example, names that are easier to pronounce are considered more likable and more trustworthy (Reber et al. 1998; Shah and Oppenheimer 2007, 2009). Presenting information using clear fonts, for example, 12-point sans serif typefaces, led to people having more confidence in their abilities to answer and correctly evaluate information relative to when that same information was presented using harder-to-read fonts, for example, small, gray, italicized, condensed font (Alter et al. 2007). Text that is simple and easy to process is also perceived to be more trustworthy and authored by someone who is more intelligent (Oppenheimer 2006).

In addition to fonts, prior work has found that the positive benefits associated with fluency can occur using virtually any method that makes information easier to process. Some of the first work done on processing fluency shows that repeated exposure of neutral stimuli can increase familiarity and fluency, leading to the same stimuli being rated more positively over time (i.e., the mere exposure effect, Zajonc 1968). Methods that have improved the visual clarity or contrast, providing simpler information displays like checklists or bulleted lists, or interventions that reduce the amount of information presented on forms, have all produced positive effects on fluency and evaluation (Bornstein 1989; Reber et al. 1998).

Improving the ease of processing can also influence cognitive processes. People comprehend more information, make faster decisions, and integrate information more frequently when processing fluency is high (Peters et al. 2007). For example, sorting information by a particular attribute, such as quality or price, can ease information processing (Johnson 2021). However, the way information is sorted can influence decision-making by increasing the salience of the attribute the alternatives are sorted on (Diehl and Zauberman 2005; Lynch and Ariely 2000; Quaschnig, Pandelaere, and Vermeir 2014). One example of how sorting can influence decision-making is in the context of school choice. Glazerman (2017) provided parents with a website sorting schools in several ways, such as by academic quality or geographic distance. When information was sorted automatically by academic performance, families were more likely to choose schools that were academically stronger but farther away. In all cases, the same information was available to parents, but sorting made it easier to see the differences in academic quality more easily. In contrast, information that is sorted by attributes less relevant (e.g., sorting alphabetically) makes the relevant information harder to process, reducing the likelihood that the information will be recalled and used in the decision process.

From an informational design and policy standpoint, improving the simplicity and ease of information processing can have a first-order impact on the choices consumers make. Reducing the informational complexity of even

basic informational materials (e.g., forms, brochures) has been shown to improve the uptake of various social welfare benefit programs (Bhargava and Manoli 2015), increase the likelihood of getting a flu vaccine (Milkman et al. 2011), and can help people choose an optimal health insurance or prescription drug plan (Kling et al. 2012). Taken together, this literature stresses that policymakers and practitioners looking to encourage a particular behavior, such as improving retirement contributions, should focus on interventions that improve fluency and simplify information on forms and account statements. A corollary to this notion is that we might also expect that disfluent retirement forms and statements, which require greater cognitive effort to process (Alter and Oppenheimer 2009), will reduce retirement contributions.

## Fluency Amplification Model: Current Empirical Evidence

The empirical evidence has largely found processing fluency to positively impact evaluations, but this may not always be the case. Much of the work looking at fluency effects have focused on positive or neutral stimuli (Albrecht and Carbon 2014). However, information is not always positive. A product may be lower in quality or higher in price relative to other options. If information containing negative or unfavorable information is made more fluent, then we may expect negative information to be easier to process, evaluate, and encode in the decision process. Albrecht and Carbon (2014) provided some of the first evidence showing that fluency effects could have potential negative consequences. They propose that fluency is a dynamic process that can amplify existing positive and negative evaluations. According to the fluency amplification model (FAM), fluent positive stimuli will be evaluated more positively, and negative stimuli evaluated more negatively, than if the same information were presented disfluently.

While this notion is intuitive, evidence in favor of this account is relatively limited. The original work by Albrecht and Carbon (2014) presents two empirical studies, only one of which supported FAM (see Landwehr and Eckmann 2020 for an excellent overview). In their first experiment with a notably small sample ( $N = 35$ ), positive stimuli that were presented more fluently were evaluated more favorably, while negative stimuli presented more fluently were evaluated more negatively. In their second experiment ( $N = 39$ ), they failed to replicate the negative amplification effects, replicating positive amplifying effects in only the highest positive category of target valence. Moreover, both studies never tested processing fluency measures, which limits evidence for the potential underlying mechanism. Landwehr and Eckmann (2020) provide direct evidence of fluency amplification via a process fluency mechanism.



Importantly, however, both papers find evidence for fluency amplification effects on outcomes involving how stimuli are perceived and evaluated. Less is known about whether FAM can also influence behavioral outcomes, particularly at scale. The notion that fluency can produce both positive and detrimental effects has implications for understanding important and substantive decision outcomes. If making information fluent and simpler can have the potential to produce backfire effects, then it is imperative to identify the conditions for when and how simplification can be leveraged by policymakers and practitioners.

Using two field experiments with two retirement providers in Mexico (one firm that is highly ranked in rate of returns and another that is lower ranked), we test whether simplification always yields positive results and encourages retirement savings contributions, or alternatively, whether there are contexts when simplification may unintentionally backfire.

We present our results as well as provide a comparison of our effects across firms. Because individuals self-select into firms, and the characteristics of individuals may be different in each firm, we correct for cross-firm selection using a procedure that builds on recently developed machine learning techniques. The procedure flexibly estimates an individual's treatment effect conditional on their observable characteristics (e.g., salary, location, gender) and then computes the difference in average treatment effects across the two firms. In constructing the firm-level averages, individuals are reweighted so that the two populations are comparable.

We also conduct several robustness checks using follow-up laboratory experiments to reduce cross-firm selection concerns and alternative accounts. We also test whether simplifying multiple paths to improve retirement savings, that is, simplifying the perceived cost of switching to a better performing firm, can harness simplification to improve consumer welfare more broadly. Finally, in experiment 2, we test a potential mechanism underlying our effects. We use a process-tracking experiment to show that our bidirectional effects are, at least partially, driven by a processing fluency mechanism. We conclude with a general discussion of the implications of improving processing fluency via simplification for theory, policy, and information design.

## FIELD EXPERIMENTS 1A AND 1B: SIMPLIFYING RETIREMENT ACCOUNT STATEMENTS IN MEXICO

For experiments 1A and 1B, we collaborated with Mexico's government regulatory commission, CONSAR (Comisión Nacional del Sistema de Ahorro para el Retiro), as well as two private retirement fund administrators known as AFOREs (Administradoras de Fondos para el Retiro). Both retirement companies had a high market

share, each having over 2 million clients. We used a representative sample from each firm and tested whether simplifying and improving the visual appeal of retirement account statement forms influenced voluntary retirement savings rates relative to the standard retirement account forms typically distributed to customers.

## Background

Mexico is a particularly compelling context for the present research. Despite having the world's 15th largest GDP and the second largest economy in Latin America, nearly 28% of those aged 65 and older are unable to retire comfortably and must continue participating in the labor force. The nearly 28% participation rate is roughly nearly double the Organisation for Economic Cooperation and Development (OECD) average and second worst of all OECD countries. While the elderly poverty rate is thought to be comprised predominantly of those in the informal labor sector who may not have the means to save for retirement at earlier points in time, retirement savings rates even among relatively privileged formal sector employees are very low. Less than 0.5% of full-time workers make a voluntary savings contribution annually (Fertig et al. 2015).

In 1997, the Mexican social security system underwent significant reforms aimed at helping people save and prepare for their retirement by providing more consistency and transparency for pension contributions. The government partnered with several private retirement funds, or AFOREs, switching from a pay-as-you-go system to a private defined contribution system. All full-time workers in Mexico are required to select one of several government-designated retirement fund administrators known as AFOREs for their mandatory retirement contributions. After selecting a company, the employer directs 6.5% of a worker's salary to their retirement account. These mandatory contributions are mostly paid by employers, with employees and the government providing a very small portion of the total amount (Mitchell 1999). Historically, the combined total of the mandatory employee contributions, employer contributions, and the accrued interest earned would equate to receiving roughly 30% of the average earned salary at the time of retirement—far less than the 70% target typically recommended by financial experts (Mitchell 1999; Munnell et al. 2011). Account holders are expected to close the gap by making additional *voluntary* contributions to their retirement account. However, less than 1% of account holders do so (Fertig et al. 2015).

AFOREs tend to have segmented markets, appealing to different customers across numerous attributes. AFOREs typically have more physical branches and sales force in particular areas, subsequently leading to greater market share in some regions versus others. AFOREs differ in the size of sales force teams and the commission structure and rates that they offer for to their respective salesforce for

bringing in new clients or increasing voluntary contributions (Hastings, Hortaçsu, and Syverson 2013; Mitchell 1999).<sup>1</sup> AFOREs also vary in terms of how long they were established, their overall market share, and which channels they use for advertising, or even if they chose to advertise at all (Hastings et al. 2013). Differences in commission rates and advertising tend to correlate with differences in the rate of returns across AFOREs.

To regularly update citizens about the status of their retirement savings accounts, the government required all providers to issue account statements roughly every 4 months. Government regulations also required account statements to be kept virtually identical across firms. Account statements provided accountholders with updated information on their current account balance, the amount saved to date, how much their employer contributed, and the amount of money accrued from interest and investment returns. Account statements also provided information on how to switch firms as well as how to make an additional, voluntary contribution to their savings providers beyond what was mandatory. Voluntary contributions could be made with any frequency or amount exceeding \$50 MXN (\$2.35 USD) and could be made through various channels indicated on the statement.

The government also required providers to disclose how assets were allocated across various investments. The government required a baseline of diversification of holdings and regulated the types of assets where providers could invest (e.g., domestic versus international holdings, stock versus bonds). However, providers naturally used different investment strategies, allocating different proportions across the various asset classes. Differences in investment performance along with the operational, advertising, as sales force commission fee costs meant that AFOREs varied on their rate of returns.

To improve transparency of rate of return information and to make it easier for citizens to compare differences across firms, the government required all AFOREs to provide a standardized rate of return comparison table on account statements and rank ordering firms from highest to lowest performing. Rankings were presented identically across providers, regardless of the firm or the firm's position on the table. Some have speculated that providing the ranking information has led to a departure of less competitive firms from the market, shrinking the overall number from 17 in

1997 to 11 firms at the start of our experiment (Calderón-Coli'n, Domínguez, and Schwartz 2010). By 2021, the number of firms available shrank to 10. It is quite typical for a firm's ranking to move up and down based on investment performance changes that naturally may occur throughout the year or across years. We collaborated with one firm that was ranked within the top two highest for expected returns and another firm that was ranked within the bottom three.

## Experimental Materials and Procedure

A representative sample of over 60,000 account holders was selected from each AFORE, and randomly assigned to receive one of five experimental conditions. Each company stratified individuals into the experimental conditions based on age group (i.e., Siefore), gender, and whether they voluntarily contributed in the 1-year period prior to the start of our experiment. Consistent with broader retirement savings trends, we observed very low baseline rates of contributions across both firms. To ensure that prior contribution rates do not result in pre-treatment differences across conditions, we also obtained individual voluntary contribution data for roughly 1 year prior to the beginning of the experiment. We find no differences in prior contribution likelihood across our treatments in either firm (see tables 2 and 3 for baseline summary statistics).

Those in the control condition received the standard account statement that was typically sent by the company. Individuals in the experimental conditions received one of four simplified forms that improved the readability and visual appeal of forms and included a behavioral nudge intervention (i.e., gain frame, loss frame, future visualization prompt, or wallet cutout). Figures 1 and 2 provide a sample of the control and simplification form variants (see web appendix B for a sample of the forms in Spanish and English). Consistent with government regulations regardless of the firm or experimental condition, all account statements contained information on their current account balance, projected monthly income at the time of retirement, the firm's absolute rate of return, and how the firm compared relative to other government approved retirement firms. Each form also provided information for how individuals could make additional, voluntary savings contributions. The four simplification variations designed for firm 1B were virtually identical to firm 1A, with the firm-specific information changed accordingly (e.g., logo, branch information, asset distribution, and website links).

Simplified redesigned statements included a thermometer on the top-left emphasizing how well the consumer was doing at saving for retirement. An indicator in the red region suggested savings were inadequate, while an indicator in the green region suggested that savings were good. Due to the historically low rates of contributions for a large majority of customers, the thermometer indicator was placed in between the yellow and red zones for everyone

1 Firms are compensated for collection of contributions, money management, record-keeping, and benefit payouts via commission charges. Larger providers leverage operational efficiencies and tend to have lower costs per client. The Mexican government permits each AFORE to design its own commission scheme (e.g., one-time up-front fees based on the contributions, annual percent of assets, annual real returns, commissions on both contribution amounts and returns). Commission rates also differ across firms, with larger providers having lower commissions. To remain profitable, some smaller providers merged with larger providers, reducing the total available firms. At the time of our experiment, there were 11 AFOREs available.

TABLE 2

BASELINE SUMMARY STATISTICS AND BALANCE CHECKS (BY TREATMENT), FIELD EXPERIMENT 1A: HIGH-RANKING

| Condition                      | <i>N</i> | Proportion Female | Prior Contributor | Siefore 4 (Age < 37) | Siefore 3 (37–45) | Siefore 2 (46–59) |
|--------------------------------|----------|-------------------|-------------------|----------------------|-------------------|-------------------|
| Standard statement (Control)   | 12,227   | 44.18%            | 1.13%             | 80.94%               | 13.66%            | 5.40%             |
| Simplification (overall)       | 50,569   | 44.15%            | 1.15%             | 80.82%               | 13.79%            | 5.39%             |
| <i>Treatment variant</i>       |          |                   |                   |                      |                   |                   |
| Gain frame                     | 12,639   | 44.25%            | 1.21%             | 80.73%               | 13.81%            | 5.47%             |
| Loss frame                     | 12,625   | 43.91%            | 1.21%             | 80.83%               | 13.79%            | 5.38%             |
| Wallet cutout                  | 12,662   | 44.00%            | 1.06%             | 80.88%               | 13.79%            | 5.33%             |
| Concrete future                | 12,643   | 44.44%            | 1.12%             | 80.86%               | 13.77%            | 5.37%             |
| Total <i>N</i>                 | 62,796   |                   |                   |                      |                   |                   |
| <b>p-value of regression F</b> |          | .927              | .755              | 1.00                 |                   |                   |

TABLE 3

BASELINE SUMMARY STATISTICS AND BALANCE CHECKS (BY TREATMENT), FIELD EXPERIMENT 1B: LOW-RANKING

| Condition                      | <i>N</i> | Proportion Female | Prior Contributor | Siefore 4 (Age < 37) | Siefore 3 (37–45) | Siefore 2 (46–59) | Siefore 1 (59 < age) |
|--------------------------------|----------|-------------------|-------------------|----------------------|-------------------|-------------------|----------------------|
| Standard statement (control)   | 13,138   | 45.03%            | 0.51%             | 77.93%               | 16.11%            | 5.40%             | 0.13%                |
| Simplification (overall)       | 51,732   | 45.64%            | 0.38%             | 78.13%               | 16.02%            | 5.39%             | 0.08%                |
| <i>Treatment variant</i>       |          |                   |                   |                      |                   |                   |                      |
| Gain frame                     | 12,961   | 45.40%            | 0.41%             | 78.17%               | 15.81%            | 5.47%             | 0.09%                |
| Loss frame                     | 12,946   | 45.10%            | 0.34%             | 78.21%               | 15.98%            | 5.38%             | 0.03%                |
| Wallet cutout                  | 12,915   | 46.67%            | 0.38%             | 78.08%               | 16.22%            | 5.33%             | 0.11%                |
| Concrete future                | 12,910   | 45.42%            | 0.41%             | 78.04%               | 16.09%            | 5.37%             | 0.10%                |
| Total <i>N</i>                 | 64,870   |                   |                   |                      |                   |                   |                      |
| <b>P-value of regression F</b> |          | .217              | .280              | .624                 |                   |                   |                      |

regardless of firm or their previous contribution history or accumulated amount. All simplification form variants included an actionable checklist for everything needed to make a voluntary contribution. The control statement, in contrast, contained the same instructions but in sentence form in a footnote that was located at the bottom of the statement. The back of the redesigned account statements contained information on expected returns for money invested voluntarily in the retirement account versus in a traditional savings account.

To ensure that our results were driven by simplification rather than one particular format alone, we strategically modified the content and appearance of the simplified forms by testing four different simplification statement variations manipulating either by varying the location of the ranking table (e.g., top of the page or bottom of the page), the way that the monetary advantages of saving via retirement fund was framed versus a traditional savings account (e.g., gain frame or loss frame), increasing the visual appeal and imagery (e.g., heightening the salience of the future with a bigger or smaller image of future outcomes), or whether a cutout memory aid (i.e., a wallet card) was included. Pairwise comparisons between interventions

reveal no differences across simplification form variants. As a result, we present our analyses first comparing each respective simplification form variant to the status quo, control form, and then combining all simplification form variants and comparing this overall simplification treatment relative to the control form.

Both field experiments commenced at the beginning of May when all forms were sent to customers, and ran until the end of September, when the next account statement was sent. The main outcome variables were (i) the proportion that made a voluntary contribution during the experimental period and (ii) the total amount contributed to their voluntary savings account. In later analyses, we also measure the likelihood of switching AFOREs as a function of experimental condition.

### Field Experiment 1A: High-Ranking Firm Results

The representative sample from firm A included 62,796 active customers (44.2% female,  $M_{\text{age}} = 32.6$ , table 2) with roughly 81% falling between the ages of 19 and 36. There were no significant pre-treatment differences between the

FIGURE 1

FIELD EXPERIMENTS 1A AND 1B, SAMPLE CONTROL ACCOUNT STATEMENTS

### STANDARD FORM FRONT SIDE (CONTROL GROUP)

The front side of the standard form includes a header with 'Afore SB(x)' and 'ESTADO DE CUENTA / DECIMO TRANSITORIO'. It contains fields for 'Periodo que comprende este Estado de Cuenta', 'Sus datos' (CURP, RFC), and 'Atención al público'. A 'Resumen general' section shows 'Saldo anterior', 'Aportaciones', 'Retiros', 'Rendimientos', 'Comisiones', and 'Saldo final'. Below this is a table for 'Ahorro para el retiro' and 'Ahorro voluntario'. At the bottom, there is a 'RENDIMIENTO BANKARIO' table and a 'RENDIMIENTO NETO' table with a pie chart showing investment instruments.

### STANDARD FORM BACK SIDE (CONTROL GROUP)

The back side of the standard form includes a header with 'SB(x)' and 'ESTADO DE CUENTA'. It features a large 'Espacio para publicidad AFORE' and 'Espacio para información CONSAR'. Below these are sections for 'RESUMEN DE MOVIMIENTOS' and 'Su Ahorro para el Retiro', followed by tables for 'Su Ahorro Voluntario/Solidario' and 'Su Ahorro para la Vivienda'. The bottom section shows 'SALDO TOTAL AL FINAL DEL PERIODO'.

control and any of the treatment conditions prior to the experimental intervention being administered, indicating that intervention and control participants were similarly distributed and balanced at baseline ( $ps > .75$ ).

Table 4 presents the results of our analyses. The four treatment interventions were not significantly different from one another ( $p = .91$ ), allowing us to present results both by the specific simplification variant and collapsed across all four simplification treatment variants. Using a z-test of proportions, we find that each simplification variant significantly improves contribution rates relative to the status quo, control form ( $ps < .022$ ). Receiving a simplified intervention statement, regardless of form variant, improved the contribution likelihood by 39% ( $z = 3.58$ ,  $p < .001$ ). The increase in contribution likelihood ranged from an improvement of 32% for the Concrete Future simplification variant to 43% for the Wallet Cutout variant (figure 3).

Receiving simplified forms also improved voluntary contribution amounts, when comparing all four

simplification variants overall ( $B = 0.026$ ,  $SE = 0.008$ ,  $p = .001$ ) and when comparing each simplification variant separately to the standard form used ( $ps < .036$ ). Relative to the control statement, receiving a simplified account statement increased total contributions by 43%. The Gain Frame form variant produced the largest relative difference, improving contribution amounts by 76% relative to the status quo, standard form. For high-ranking firm customers, simplifying retirement account forms improves both the likelihood of making a voluntary savings contribution and the total amount contributed to voluntary retirement savings accounts.

### Field Experiment 1B: Low-Ranking Firm Results

The representative sample from firm B included 64,870 active customers (45.5% female,  $M_{age} = 33.7$ , table 3) with 78% falling between the ages of 19 and 36. In total, the control condition had 13,078 individuals, while the four simplification form variants included 51,566 individuals



FIGURE 2

FIELD EXPERIMENTS 1A AND 1B, SAMPLE SIMPLIFICATION ACCOUNT STATEMENTS

### SIMPLIFICATION FRONT SIDE

### SIMPLIFICATION BACK SIDE

with roughly 12,900 individuals per each simplification variation. Like firm 1A, we observe no significant pre-treatment differences across baseline covariates, indicating that experimental groups were similarly distributed and sufficiently balanced at the start of our experimental period ( $p$ s > .20).

Table 5 presents the results of our analyses. Once again, we find that the four treatment interventions were not significantly different from one another ( $p$  = .85), thus allowing us to present results both by each simplification variant and comparing simplification overall by collapsing across the four simplification treatment variants. In contrast to high-ranking firm customers (firm 1A), we find that for low-ranking firm customers, the simplified forms *decreased* the likelihood of contributing relative to those receiving the standard account statement. The Loss Frame simplification variant significantly lowered the proportion making a voluntary retirement savings contribution ( $z$  = -2.28,  $p$  = .023), while the Wallet Cutout intervention marginally lowered voluntary contribution rates,  $z$  = -1.82,  $p$  = .069. Though the Gain Frame and Concrete Future interventions did not significantly lower contribution likelihood,

both had lower contribution rates relative to sending out the standard, control form ( $p$  > .131).

Collapsing across simplification form variants results in a significantly lower likelihood of contributing as compared to the control ( $z$  = -2.37,  $p$  = .018). As shown in figure 4, sending account holders a simplified form results in a 30% decrease in voluntary savings contributions relative to using the standard account statement. Simplification also produced backfire effects on total contribution amounts, significantly lowering savings contributions by roughly 4%,  $B$  = -0.010,  $SE$  = 0.005,  $p$  = .035. The loss frame form variation had the greatest backfire, significantly decreasing contribution likelihood by nearly 37% ( $p$  = .023) and contribution amounts by 37% ( $p$  = .026) relative to sending out the more complex, standard account statement.

### Simplification Effects on the Likelihood to Switch Firms

From a welfare standpoint, the decision to make a voluntary retirement savings contribution may not be optimal for

**TABLE 4**  
EFFECTIVENESS OF SIMPLIFICATION, FIELD EXPERIMENT 1A: HIGH-RANKING

| Contribution Likelihood      |         |           |                   |       |           |
|------------------------------|---------|-----------|-------------------|-------|-----------|
| Condition                    | Mean    | SD        | t(SE)             | p     | Cohen's d |
| Standard statement (control) | 0.93%   | .096      | —                 |       |           |
| Simplification (overall)     | 1.29%   | .113      | 3.58***<br>(.001) | <.001 | 0.033     |
| <i>Treatment variant</i>     |         |           |                   |       |           |
| Gain frame                   | 1.29%   | .113      | 2.69***<br>(.001) | .007  | 0.035     |
| Loss frame                   | 1.31%   | .114      | 2.81***<br>(.001) | .005  | 0.036     |
| Wallet cutout                | 1.33%   | .115      | 3.00***<br>(.001) | .003  | 0.038     |
| Concrete future              | 1.23%   | .110      | 2.30***<br>(.001) | .021  | 0.029     |
| N                            | 62,796  |           |                   |       |           |
| Contribution Amount (Pesos)  |         |           |                   |       |           |
| Condition                    | Mean    | SD        | t(SE)             | p     | Cohen's d |
| Standard statement (Control) | \$38.85 | \$692.23  | —                 |       |           |
| Simplification (Overall)     | \$55.53 | \$1192.82 | 3.28***<br>(.008) | .001  | .033      |
| <i>Treatment variant</i>     |         |           |                   |       |           |
| Gain frame                   | \$68.37 | \$1906.76 | 2.46***<br>(.010) | .014  | .035      |
| Loss frame                   | \$47.97 | \$667.61  | 2.66***<br>(.010) | .008  | .036      |
| Wallet cutout                | \$53.86 | \$932.55  | 2.76***<br>(.010) | .006  | .038      |
| Concrete Future              | \$51.93 | \$860.47  | 2.09***<br>(.010) | .036  | .029      |
| N                            | 62,796  |           |                   |       |           |

NOTE.—Standard errors are reported in parentheses. Coefficients significant at the 10%, 5%, and 1% levels are marked with a \*, \*\*, and \*\*\*, respectively.

all individuals. If people are facing budget constraints, then attending to more pressing financial concerns is rational. If the number of people facing a budget constraint is correlated with people who select into the lower-ranked firm, then observing a lower savings rate when the form is simplified may be beneficial. Alternatively, if there are outside options that offer higher interest rates relative to the retirement savings provider, then it would also be financially prudent to divert resources to options offering higher returns, regardless of where the retirement firm is ranked. In each instance, observing significantly lower voluntary contributions does not necessarily mean that form simplification is detrimental toward the overall goal of improving retirement savings. Since we only have access to contributions to their retirement firm, we are unable to determine whether simplification influenced outside investment allocations or the tradeoff between meeting current needs versus saving for retirement.

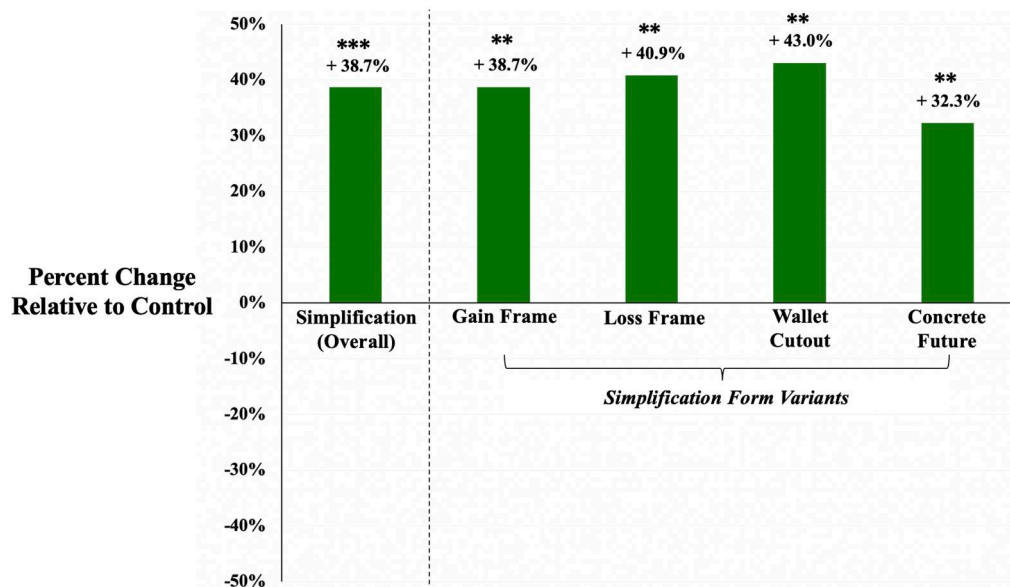
Aside from voluntary savings, consumers are also able to switch firms at any point without incurring any potential transfer or activation fees. Regardless of whether people are financially constrained or whether higher performing

outside options exist, choosing a firm with a relatively higher return rate is optimal. If simplifying the form increases the fluency of processing negative information, motivating people to switch to a higher performing alternative relative to their firm, then simplification might still help people improve their retirement savings. Testing whether simplification could be utilized in a way to motivate individuals to switch firms is of particular interest in Mexico given that more than 99% of active accountholders only make mandatory contributions.

To test switching likelihood, we acquired data from both AFOREs indicating whether customers in our sample switched to one of the other 10 AFOREs available (Yes/No). Though each firm did not have information regarding the specific firm chosen post-switch, we are able to make certain inferences based on the ranking information. From a probability standpoint, for higher-ranking firm customers (firm 1A), switching at random would yield an 80% likelihood of choosing a firm with a lower rate of returns. For lower-ranking firm customers (firm 1B), switching at random would lead to an 80% likelihood of choosing a firm with a higher rate of returns. Therefore, the baseline

FIGURE 3

HIGH-RANKING FIRM, PERCENTAGE CHANGE IN CONTRIBUTION LIKELIHOOD



assumption is that switching firms is most likely a financially beneficial decision for lower-ranked firm customers but detrimental for those in the highly ranked firm. If the simplification intervention encouraged lower-ranking firm customers to switch to a different firm, then there still might be a positive welfare effect overall for form simplification even if it comes at a cost to the firm (particularly as the probability of switching to a better performing firm is high even if those at the low-ranking firm choose at random).

Using a *z*-test of proportions, we investigated whether simplification influenced the likelihood to switch to a different firm relative to the standard account statement. Table 6 and figure 5 reflect the results of simplification intervention relative to the standard form in each firm. Form simplification does not significantly influence the likelihood to switch firms for high-ranking firm individuals,  $z = -0.944$ ,  $p = .345$ . There was also no significant difference in the proportion switching for low-ranking firm individuals between those who received a simplified form versus those who received the standard account statement,  $z = -0.643$ ,  $p = .520$ . Probing further, none of the simplification form variants led to a significant change in switching likelihood in either of the two firms ( $ps > .13$ ).

The low incidence of switching that we observe is consistent with broader trends. Nearly 90% of consumers never change firms over the course of their lifetime. Historically, switching has not been indicative of individuals trying to obtain higher returns. For the 10% of those who switch

firms, roughly two-thirds move to a firm with a lower rate of return. Combining the outcomes on switching likelihood along with results on voluntary contribution rates suggests that for high-ranking firm customers, reducing the cognitive complexity of retirement statement forms may improve contribution rates, benefiting retirement savings overall. However, simplifying retirement account statement information may not be as uniformly positive for those in a low-ranking firm.

### Comparing Results Across Firms

Although individuals within each firm were randomly assigned to either the control or one of our intervention groups, individuals self-select into AFOREs, leaving open the very real possibility that our estimated treatment effects are due to endogenous group formation. Effectively, our treatment may have heterogeneous effects across our two firms due to differences in the kinds of people who join each firm, rather than the simplification treatment itself.

Typically, cross-firm comparisons pose significant challenges making this type of analysis very difficult. We address the issue of cross-firm selection using two different approaches: a newly developed machine learning technique as well as follow-up laboratory experiments. The intuitive idea behind our machine learning procedure is that for each customer in firm 1A, we will find a customer in firm 1B who has similar characteristics, compute the difference in treatment effects for these two customers, and then average

**TABLE 5**  
EFFECTIVENESS OF SIMPLIFICATION, FIELD EXPERIMENT 1B: LOW-RANKING

| Condition                    | Contribution Likelihood     |           |                    |          |                  |
|------------------------------|-----------------------------|-----------|--------------------|----------|------------------|
|                              | Mean                        | SD        | <i>t</i> (SE)      | <i>p</i> | Cohen's <i>d</i> |
| Standard statement (Control) | 0.46%                       | .067      | —                  |          |                  |
| Simplification (overall)     | 0.32%                       | .057      | −2.13***<br>(.001) | .033     | .023             |
| <i>Treatment variant</i>     |                             |           |                    |          |                  |
| Gain frame                   | 0.34%                       | .058      | −1.50<br>(.001)    | .133     | .019             |
| Loss frame                   | 0.29%                       | .053      | −2.27***<br>(.001) | .023     | .028             |
| Wallet cutout                | 0.32%                       | .056      | −1.81***<br>(.001) | .070     | .022             |
| Concrete future              | 0.34%                       | .058      | −1.48<br>(.001)    | .138     | .018             |
| <i>N</i>                     | 64,870                      |           |                    |          |                  |
| Condition                    | Contribution Amount (Pesos) |           |                    |          |                  |
|                              | Mean                        | SD        | <i>t</i> (SE)      | <i>p</i> | Cohen's <i>d</i> |
| Standard statement (Control) | \$12.01                     | \$692.23  | —                  |          |                  |
| Simplification (overall)     | \$11.55                     | \$1192.82 | −2.11***<br>(.001) | .001     | .033             |
| <i>Treatment variant</i>     |                             |           |                    |          |                  |
| Gain frame                   | \$9.54                      | \$197.41  | 2.46***<br>(.010)  | .014     | .035             |
| Loss frame                   | \$7.56                      | \$167.34  | 2.66***<br>(.010)  | .008     | .036             |
| Wallet cutout                | \$9.00                      | \$227.31  | 2.76***<br>(.010)  | .006     | .038             |
| Concrete future              | \$20.11                     | \$1080.45 | 2.09***<br>(.010)  | .036     | .029             |
| <i>N</i>                     | 62,796                      |           |                    |          |                  |

NOTE.—Standard errors are reported in parentheses. Coefficients significant at the 10%, 5%, and 1% levels are marked with a \*, \*\*, and \*\*\*, respectively.

over the differences. The procedure builds on work for estimating conditional average treatment effects using double-debiased machine learning by Chernozhukov et al. (2018) and Semenova and Chernozhukov (2021). We will then present a summary of our laboratory results in the next section, with greater detail provided in web appendixes E–G.

We first start by examining how the populations of the firm differ along the observable customer characteristics to which we have access. In web appendix C, we provide the firm-level distributions of the demographic characteristics across firms. We find that the age distribution is similar across firms, with firm 1B having a marginally older population. The fraction of female customers is small but significantly different (Proportion Female Firm 1A = 44.16%, Proportion Female Firm 1B = 45.52%,  $p < .001$ ). Both firms have very few customers who contribute voluntarily (i.e., their voluntary savings balance is zero). However, we do find that the customers in firm 1A have higher overall account balances (e.g., higher mandatory contributions). This could arise due to differences in income across the firms, and/or differences in amount of time that people

were employed full-time (regardless of employer). Consistent with this account, we find a difference in where customers reside across firms. Firm 1A has more customers residing in the state of Mexico. This difference may help explain the differences in account balances across the two firms, as the state of Mexico typically has a higher population of white collar and government employees relative to other states. We also observe firm 1A's customers to have significantly higher daily salaries, though this measure is less reliable in our dataset.

Next, we investigated the interaction of our treatment interventions as a function of the firm itself. A standard approach is to use regression models to compare the treatment effects by estimating firm–treatment interactions. Given the differences in population characteristics across firms, some of the differences in firm-level treatment could arise from self-selection. In web appendix C, we show that under a standard set of independence assumptions, computing individual-level treatment effects and re-weighting firm populations to make them comparable can correct for potential selection bias. To implement the approach, we



FIGURE 4

LOW-RANKING FIRM, PERCENTAGE CHANGE IN CONTRIBUTION LIKELIHOOD

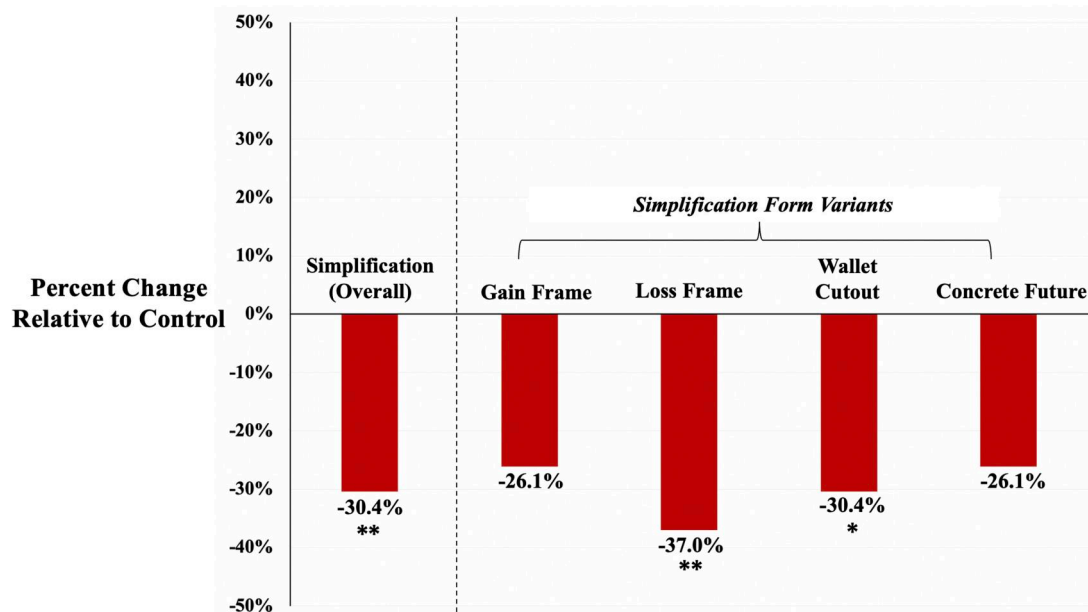


TABLE 6

PROPORTION SWITCHING AS A FUNCTION OF SIMPLIFICATION INTERVENTION

| Condition                    | High-ranked firm (Firm 1A) |         |      | Low-ranked firm (Firm 1B) |         |      |
|------------------------------|----------------------------|---------|------|---------------------------|---------|------|
|                              | Mean                       | z-score | p    | Mean                      | z-score | p    |
| Standard statement (control) | 1.03%                      | —       | —    | 2.66%                     | —       | —    |
| Simplification (overall)     | 0.93%                      | −.944   | .345 | 2.65%                     | −.643   | .520 |
| <i>Treatment variant</i>     |                            |         |      |                           |         |      |
| Gain frame                   | 1.10%                      | .596    | .551 | 2.57%                     | −.480   | .632 |
| Loss frame                   | 0.84%                      | −1.50   | .134 | 2.78%                     | .579    | .562 |
| Wallet cutout                | 0.88%                      | −1.18   | .236 | 2.50%                     | −.823   | .407 |
| Concrete future              | 0.89%                      | −1.04   | .298 | 2.40%                     | −1.35   | .177 |
| N                            | 62,796                     |         |      | 64,870                    |         |      |

NOTE.—Coefficients significant at the 10%, 5%, and 1% are marked with a \*, \*\*, and \*\*\*, respectively.

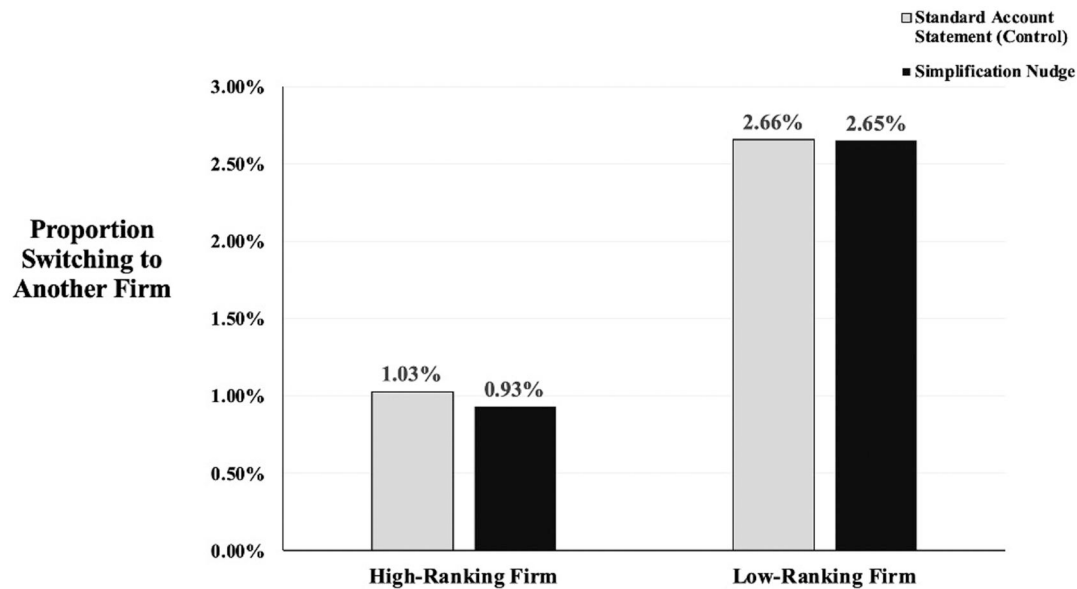
use a machine learning technique that allows us to flexibly estimate treatment effects for each individual, conditional on the individual's observable characteristics. Effectively, for each customer in firm 1A, we find a customer in firm 1B who has similar characteristics, compute the difference in treatment effects for each customer respectively, and then average over the differences.

Our approach builds on Chernozhukov et al. (2018) and Semenova and Chernozhukov (2021) and uses double-debiased machine learning to estimate conditional average treatment effects in four steps. First, for each treatment

arm, we run a random forest to compute a prediction of how contribution likelihood varies with our observables, that is, firm, gender, age, state of residence, prior contribution, and salary. Second, we compute a debiased estimate of each person's treatment effect following Semenova and Chernozhukov (2021). Third, we compute a set of weights that will reweight individuals in firm 1B so the population distribution of individuals in firm 1B is the same as firm 1A. For example, if the proportion of women in firm 1A is 60%, but firm 1B is 45%, then we would reweight the women in firm 1B to look like firm 1A. The fourth and

FIGURE 5

PROPORTION SWITCHING AS A FUNCTION OF SIMPLIFICATION INTERVENTION



final step is to run a regression of the weighted debiased treatment effect on an indicator for firm 1A.

We present the results of our regression and double machine learning models comparing the effectiveness of treatment interventions across firms in [web appendix C](#). We find that the results are identical to those shown in [tables 4 and 5](#). The redesigned simplified form led to a significant increase in contribution in firm 1A, and significant decrease in firm 1B. Because the firm populations have been reweighted, this difference shows an “apples-to-apples” comparison across the firms. The differences between the treatments across firms 1A and 1B are positive and significant, showing that, when population differences are adjusted for, the new forms increase contributions significantly more at firm 1A than firm 1B. We also present the estimated treatment effects computed from the double machine learning method, computing the average of the individual-level treatment effects without re-weighting. The results of the machine learning estimation are largely comparable to those of the regression.

In summary, even though the rank of the firm was provided identically in the standard account statement and the four intervention conditions across both firms, we observe a significant difference in contribution rates as a function of the firm. Next, we use a more controlled, laboratory environment approach to reduce potential endogeneity concerns using three laboratory experiments.

## Robustness of Main Results: Follow-Up Experiments

The results from the field experiment raise a few key issues. While the field offers a way to study our effects in the real environment of participants, field experiments make it difficult to control all sources of extraneous variation ([Blanchard et al. 2022](#)). First, there were many differences between the baseline, control forms, and the treatment forms beyond simplification. It is difficult to pinpoint which feature(s) caused the difference between the two statements in the field. Second, the form that was being sent was new for customers. Effectively, the novelty of the redesigned forms could have accounted for outcome differences rather than simplification.

Investigating contexts where the form is novel but where the information being simplified highlights positive or negative information can help rule out alternative possibilities. We conduct several robustness checks to ensure that our main result is robust. We also use these tests to help narrow down the psychological process underlying our effects and to determine whether simplification improves the efficiency of information processing.

*Does Form Simplification Increase Processing Fluency?* One of our main assumptions is that the simplified forms are in fact easier and simpler for people to process. The field setting made it difficult to verify whether

this was the case, so we used a follow-up laboratory experiment ( $N = 302$ ) to verify this assumption directly. We used an adapted version of the standard (control) and treatment account statements used in Mexico ([web appendix E](#)). To test for information processing fluency, we asked participants to rate the extent to which they thought the form was simple and easy to understand (seven-point Likert scale: 1 = Strongly Disagree, 7 = Strongly Agree). We measured information comprehension by testing whether participants accurately recalled the firm's ranking on the rate of return table. The redesigned form was rated as significantly more simple and easy to understand relative to the standard account statement ( $M_{\text{treatment form}} = 5.62$ ,  $SD_{\text{treatment form}} = 1.14$ ,  $M_{\text{standard form}} = 5.26$ ,  $SD_{\text{standard form}} = 1.43$ ,  $t(298) = 2.41$ ,  $p = .017$ ). Participants also recalled their firm's rank more accurately,  $z(300) = 7.51$ ,  $p < .001$ . Those who received a simplified treatment form accurately recalled their firm's rank 38.4% of the time, compared to only 3.3% in the control condition.

*Does Simplification Shift the Relative Importance of Ranking Information?* A natural consequence of form simplification is that it may shift the influence and relative importance of the information that remains. As a result, form simplification may shift the perceived importance of firm ranking information, affecting the decision to contribute to their savings account.

We used a follow-up laboratory experiment ( $N = 401$ ) to test this possibility, once again adapting the standard (control) and treatment account statements used in Mexico (see [web appendix F](#) for full experimental details). Participants either saw the simplified form or the status quo, more complex form. Both forms included the same ranking table that ranked providers on the average rate of returns, ordering providers from highest average rate of return to lowest rate of return. All participants were told that they were in the firm ranked last with a rate of return of 3.27%, with the table and ranking position appearing identically across the two conditions.

To assess whether simplification influenced the perceived importance of firm ranking information, we asked participants to rate the perceived importance of five different attributes/factors when making the decision to make a retirement contribution to the firm (rating each attribute from 0 to 100 with the total sum equaling 100; 0 = not at all a factor, 100 = the only factor considered): (i) Customer service, (ii) Operational effectiveness, (iii) Firm Ranking in Rate of Returns, (iv) Company Location, and (v) Firm asset allocation (i.e., how the firm had chosen to allocate and invest money across various assets). Compared to the control, those receiving the simplified form rated firm ranking information as significantly more important ( $p = .008$ ). Simplification also significantly reduced the perceived importance of customer service ( $p = .019$ ) and operational effectiveness ( $p = .001$ ). Effectively,

simplification influences the weight and importance of the firm ranking table, affecting the decision to contribute.

*Follow-up Experiment: Simplifying Perceived Switching Costs.* A takeaway from the two field experiments is that simplification may produce unintended maleficent effects: low-ranking firm customers are less likely to make a retirement contribution while also showing no change in the likelihood to switch to a better performing firm. However, there may be an alternative consideration. If simplification leads to a more accurate view of where a consumer's firm ranks relative to other firms, but information on how to switch providers is not made simple, then consumers may only respond by reducing contributions to their firm. If simplification can make it easier to know how to switch to a better performing alternative, then simplification can still be an efficient, low-cost tool for improving consumer welfare.

Using a follow-up experiment ( $N = 565$ ), we adapted retirement account statements varying whether the ranking information was simplified and whether firm switching information was simplified (see [web appendix G](#) for stimuli and experimental details). We measure four outcomes of interest: rank recall, voluntary contribution likelihood, motivation to switch firms, and whether participants actively made the choice to switch firms ("Which provider are you switching to? Choose 'Stay with [Firm name]' if you are not interested in switching right now."). Choosing to "Stay with [Firm name]" was coded as 0, while choosing any other firm indicated actively switching firms. Given that we were interested in testing whether simplification could help lower the perceived costs of switching to a better performing firm and subsequently improve retirement savings via improving returns for mandatory contributions, all participants were in the lowest ranked firm. The other available providers were listed in rank order from highest to lowest rate of return, though neither the rank nor rate of return was provided to participants when indicating their choice.

Sorting by rank order led to a significantly greater likelihood of recalling the firm rank ( $p = .003$ ) and led to a significantly lower likelihood of making a voluntary contribution ( $p < .001$ ). However, when both ranking information and switching information were simplified, consumers were more motivated to switch firms ( $p = .023$ ) and were more likely to switch to a top-performing firm ( $p = .042$ ). Sorting firms by rank and presenting switching information via a simple checklist format significantly increased switching likelihood (81.1%), with 70.7% of switchers choosing one of the top two firms. Sorting firms by rank but presenting information via a paragraph format led to only 66.2% switching, with 73.9% choosing top two firms.

## Discussion

The field setting provides an externally valid examination of when simplification may produce both beneficent and maleficent effects. However, there are numerous

differences between the two firms. Using follow-up robustness experiments, we show that simplification makes information easier to comprehend, improves processing fluency, and shifts the perceived importance of information that consumers use when making decisions. Simplification can still be useful regardless of where firms are ranked, as long as switching information is also simplified.

However, it is still unclear what is driving the differences in outcomes across firms. First, there are clear differences in the type of information that is provided in the simplified treatment statements relative to the standard statement. Providing a cleaner test of the effects of simplification of information while holding the content of the form constant, for example, by simplifying only the ranking information, would allow for greater internal validity of what is driving heterogeneous treatment responses. It is also possible that the effects are driven by the types of people who self-select into one firm or the other. While we control this possibility via double-debiased machine learning, endogeneity concerns remain.

Second, it is possible that the differences in response were due to novelty. The redesigned forms were different relative to what individuals typically receive. People may have paid greater attention to the novel form, making them more likely to notice their firm's rank.

Alternatively, simplification may operate by improving the efficacy of information extraction, making it easier to process and accurately recall information. If simplification improves the efficiency with which positive or negative information is extracted, then simplification will produce heterogeneous, amplifying effects. If simplification improves the processing fluency and information comprehension, making it easier to recall positive information, for example, the firm is highly ranked, then voluntary savings contributions will increase. If simplification improves processing fluency and comprehension, making it easier to recall negative information, then voluntary savings contributions will decrease. Testing these potential mechanisms in a laboratory setting, where participants are randomized into firms and where the novelty and content of the forms is controlled, would provide insight into the potential forces driving our results.

## EXPERIMENT 2: INFORMATION PROCESSING

We argue that simplification leads to more accurate information processing, making it easier to see where their firm is ranked. Consumers in a high-ranked firm will be more likely to recall that their firm is ranked highly, while those in a low-ranked firm will recall that their firm is ranked low. Thus, information comprehension and processing will drive the relationship between simplification and contribution likelihood. To ensure that the content and

novelty of the forms is constant, we test whether simplifying just the ranking information influences behavior.

Prior research has shown that sorting information on attributes can influence information acquisition and ease of processing (Bettman and Kakkar 1977; Quaschnig et al. 2014; Glazerman et al. 2020). Simplifying information and improving the ease of processing information have been found to improve comprehension, even while decreasing the amount of time spent reading information (Peters et al. 2007; Sanchez and Jaeger 2015). We leverage this work and test whether sorting information by rank is easier to process via lowering reading time and improving recall accuracy relative to sorting information alphabetically. We also test whether improving information processing and comprehension, that is, recalling firm rank, drives differences in contribution rates across high and low-ranked firms.

### Participants

We recruited 725 paid participants using Prolific's online survey sample ( $M_{\text{age}} = 26.5$ ,  $SD_{\text{age}} = 8.0$ ; Proportion Male = 49.7%, Proportion Female = 48.6%, Proportion Non-binary/Gender Queer = 1.7%). All participants were U.S. citizens and over the age of 18.

### Experimental Procedure and Dependent Measures

We design a process-tracking experiment using Pavlovia software ( $N = 725$ ) using a 2 (rank simplification: sorting by rank or sorting alphabetically)  $\times$  2 (firm rank: high or low rank) between subjects design (see web appendix H for stimuli and procedural details). Participants were told that they would be making financial choices after reading a retirement account statement from their retirement savings provider. They were informed that every month they were required to make a 6.5% contribution from their paycheck that went directly into their retirement account. Aside from this mandatory amount, they could also choose to make an additional voluntary contribution if they so desired.

Participants were instructed to view the front and back sides of the retirement account statement. Participants were randomly assigned to one of four conditions, manipulating (a) whether the retirement firm was ranked first or last relative to other firms in terms of the rate of returns, and (b) whether the table providing their rank was sorted alphabetically (more complex) or rank ordered from first to last (simplified). For alphabetical ordering, the retirement firm (firm P) was exactly halfway down relative to the other 10 firms, with 5 firms falling earlier in the alphabet and 5 firms falling later in the alphabet.

To measure visual attention, participants viewed the front and back pages of the retirement account statements using the process-tracking software Pavlovia ([www.pavlovia.org](http://www.pavlovia.org)). Pavlovia provides a way to measure the time spent



viewing specific elements of the form as well as overall reading, affording the ability to test relative visual attention via the proportion of time spent looking at one area of interest relative to time spent overall. For consistency with our field data, we designed the front and back of the statement to be roughly identical to the simplification variants used in the field experiment. Each page contained three clickable elements. On the front page of the statement, participants could click on (i) a savings thermometer to learn more about their financial health, (ii) a chart comparing interest accrued over time if they made a savings contribution, and (iii) steps on how to make a voluntary savings contribution. On the back page, participants could click on (i) the ranking table, comparing the rate of return across firms (our independent variable), (ii) helpful savings tips, and (iii) how saving with their firm compared to saving with the bank or not saving at all. For consistency across conditions, the front page and elements (ii) and (iii) on the back page were identical across conditions; thus, five out of the six elements were held constant with only the ranking table varying across conditions.

After reading the account statement, participants were told that they had received a \$1000 tax return. Participants were first asked to recall where their retirement company ranked in terms of rate of return (1st to 11th, or “Do Not Know”). Rank recall provided a way to test information processing and comprehension, that is, our second mediator of interest. Next, participants were asked about their contribution likelihood: “How likely are you to make a voluntary contribution to your retirement savings account?” (seven-point Likert: 1 = Extremely Unlikely, 7 = Extremely Likely). Finally, similar to [Albrecht and Carbon \(2014\)](#), we measured evaluation valence by asking participants: “How do you evaluate your retirement savings provider?” (seven-point Likert: 1 = Extremely Negatively, 7 = Extremely Positively).

## Results and Discussion

### *Information Processing and Comprehension.*

Consistent with past findings showing that presenting information in a more fluent manner improves information processing ([Peters et al. 2007](#); [Sanchez and Jaeger 2015](#)), we find that sorting firms by rank order significantly lowers reading time relative to sorting firms alphabetically, ( $M_{\text{rank order}} = 10.01$  seconds,  $SD_{\text{rank order}} = 7.83$ ,  $M_{\text{alphabetical order}} = 11.53$  seconds,  $SD_{\text{alphabetical order}} = 9.39$ ,  $t(723) = 2.37$ ,  $p = .018$ ).

We hypothesize that making the ranking table easier to read via sorting could make the firm’s ranking more salient increasing rank recall and recall accuracy. [Figure 6](#) presents the results with respect to recall. Less than 10% of participants in each condition reported that they did not know where the firm was ranked (Low-Rank Alphabetical = 6.1%; Low-Rank Rank Ordered = 10.0%;

High-Rank Alphabetical = 7.7%; High-Rank Rank Ordered = 3.4%). For those who answered the recall question, there was a significant main effect of firm rank and ranking table format, as well as significant interaction effect (firm rank:  $F(1, 665) = 553.4$ ,  $p < .001$ ; Ranking table format:  $F(1, 665) = 4.95$ ,  $p = .026$ ; firm rank  $\times$  table format:  $F(1, 665) = 97.93$ ,  $p < .001$ ). Those in the low-ranked firm were significantly more likely to recall their firm was lower ranked than when the ranking information was sorted alphabetically ( $B = 2.47$ ,  $SE = 0.287$ ,  $t(672) = 8.61$ ,  $p < .001$ ).

Next, we compared recall accuracy for those who responded to the firm ranking question (i.e., they did not respond with “I do not know”). Recall accuracy (0 = Not Accurate, 1 = Accurate) was significantly influenced by the main effect of firm rank, table format, as well as a significant interaction of firm rank by table format (firm rank:  $B = 2.08$ ,  $SE = 0.306$ ,  $\chi^2 = 46.1$ ,  $p < .001$ ; tablet format:  $B = 2.54$ ,  $SE = 0.305$ ,  $\chi^2 = 69.8$ ,  $p < .001$ ; Firm Rank by table format:  $B = -0.81$ ,  $SE = 0.399$ ,  $\chi^2 = 4.11$ ,  $p = .043$ ). When including all participants, we still find recall accuracy to be significantly influenced by the main effect of firm rank and table format, and the interaction between firm rank and table format (firm rank:  $\beta = 1.62$ , Robust  $SE = 0.185$ ,  $z = 8.73$ ,  $p < .001$ ; Ranking tablet format:  $\beta = 2.00$ , Robust  $SE = 0.185$ ,  $z = 10.80$ ,  $p < .001$ ). Both those in the low-ranked firm and those in a high-ranking firm were significantly more likely to recall that their firm was lower ranked than when the ranking information was sorted alphabetically (Low-ranked firm:  $\beta = 2.37$ , Robust  $SE = 0.299$ ,  $z = 7.92$ ,  $p < .001$ ; High-ranked firm:  $\beta = 1.72$ , Robust  $SE = 0.244$ ,  $z = 7.04$ ,  $p < .001$ ).

For those in the low-ranking firm, 59.4% accurately recalled their firm rank in the simplified, rank-ordered condition relative to just 10.3% in the more complex, alphabetically sorted condition. We observe the same positive effect for participants in the high-ranking firm: those who saw the table sorted by rank order were significantly more likely to recall their firm accurately, with an accuracy rate of 83.9% relative to 49.7% for those who received the ranking table sorted alphabetically.

In addition to information recall, we also tested reading time and proportion of attention spent viewing ranking information (please see [web appendix H](#) for the details of these analyses). Consistent with simplification improving processing fluency, we find that simplification via rank ordering leads to decreased reading time, while increasing the proportion of time spent viewing ranking information overall. Taken together, we show that simplification decreases reading time but increases information comprehension, visual attention placed on ranking information relative to other aspects, and recall accuracy.

### *Contribution Likelihood and Evaluation Valence.*

[Figure 7](#) depicts the effect of firm rank and table format on

FIGURE 6

RANKING RECALL AS A FUNCTION OF FIRM RANK AND TABLE FORMAT

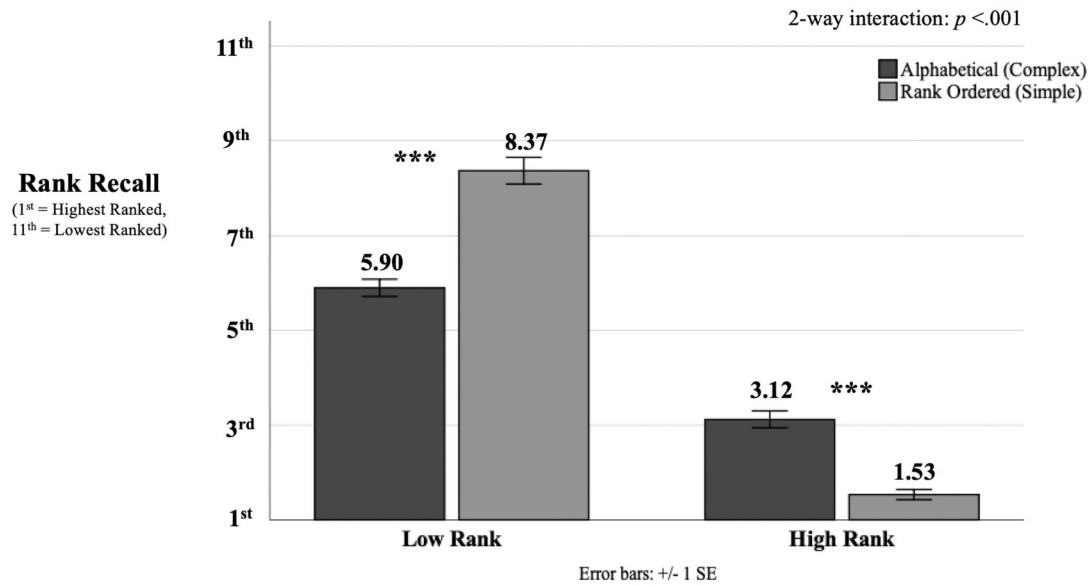
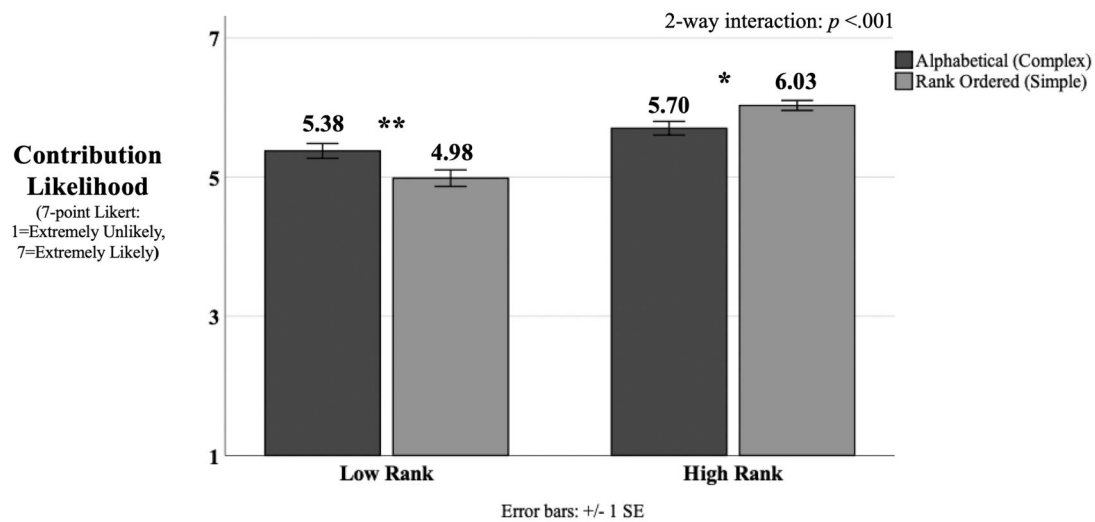


FIGURE 7

CONTRIBUTION LIKELIHOOD AS A FUNCTION OF FIRM RANK AND TABLE FORMAT



the likelihood of contributing. There was a significant main effect of firm ranking on the likelihood of making a retirement savings contribution,  $F(1, 718) = 45.04$ ,  $p < .001$ , Eta-squared = .059. There was no main effect of table format alone on contribution likelihood; however, there was

also a significant interaction effect between our two independent variables (table format:  $F(1, 718) = 0.96$ ,  $p = .756$ ; firm rank  $\times$  table format:  $F(1, 718) = 12.38$ ,  $p < .001$ , Eta-squared = .017). Planned contrasts reveal that for those in the high-ranking firm, presenting information in a simpler

manner via rank ordering significantly increased contribution likelihood relative to when the same information was presented in a more complex manner via sorting firms alphabetically ( $B = 0.327$ ,  $SE = 0.145$ ,  $t(714) = 2.25$ ,  $p = .024$ ). For those in a low-ranking firm, presenting ranking information in rank-order significantly decreased contribution likelihood ( $B = -0.391$ ,  $SE = 0.143$ ,  $t(714) = -2.72$ ,  $p = .007$ ). Evaluation valence is only marginally affected by the interaction between our two main effects ( $p = .093$ ). Those in a low-ranked firm have marginally lower evaluations of their firm when the table is rank-order versus presented alphabetically ( $p = .082$ ); however, we observe no differences of the table format for high-ranking firm customers ( $p = .509$ ).

**Mediation Analysis.** We tested a moderated mediation pathway, investigating whether information processing and comprehension (i.e., ranking recall) drove the relationship between ranking simplification and contribution likelihood, and whether the direction of this mediation pathway is moderated by whether the firm is ranked high or low. To test for this mediated moderation structure, we conducted an analysis using Hayes (2017) PROCESS module (Model 7; 10,000 bootstrap samples). The dependent variable was contribution likelihood.

Table format (alphabetical versus rank ordered), firm rank (high- vs. low-rank), and their interactions were the independent variables, while ranking recall was the mediation process variable in our model.

For our mediation analyses, we followed Zhao et al. (2010), analyzing the indirect effects of ranking simplification and firm rank on contribution likelihood via our proposed mediation variable. The results show that rank recall is a significant mediator on the effect of simplification on contribution likelihood for high and low-ranking firms ( $a_i \times b_i = -0.4993$ , 95% CI =  $[-0.4914, -0.2490]$  and  $a_i \times b_i = 0.5007$ , 95% CI =  $[0.1563, 0.3147]$ ). The differences between the conditional indirect effects for high and low rank are also significant (the indices of moderated mediation between high rank and low rank firm is 0.5964, 95% CI =  $[0.4362, 0.7731]$ ). This result provides support that the moderating effect of firm rank on the effect of ranking simplification is mediated by ranking recall (i.e., information comprehension).

Replicating our field experiment results, we find that simplifying information has a heterogeneous effect on voluntary savings contributions. Those in higher-ranked firms are significantly more likely to save for retirement, while those in lower-ranked firms are significantly less likely to save for retirement. Moreover, we find direct evidence that making the account statement simpler and easier to process via rank ordering leads individuals to recall their firm rank more accurately, affecting the likelihood of making a savings contribution. Simplification has a beneficial effect for those in higher-ranked firms, making it easier to see that

their firm is ranked highly, significantly increasing retirement savings. However, simplification detrimentally affects those lower-ranked firms by making it easier to see that their firm is ranked last, thereby decreasing both retirement savings and lowering firm perceptions.

## GENERAL DISCUSSION

Considerable research suggests making information simpler is better. Simplification improves the efficiency of information extraction and lowers psychological frictions, leading to its popularity with policymakers and practitioners worldwide. Yet, it is not clear when and how simplification can be utilized most effectively, or if there are contexts where simplification may produce unintended maleficent effects.

Using two field experiments designed to improve retirement savings in Mexico, along with follow-up laboratory experiments, we find that the effectiveness of simplifying basic forms for encouraging voluntary retirement savings is highly dependent on information valence. For the higher-ranking firm, simplification increased the likelihood of consumers recalling that their firm was ranked highly, increasing consumer confidence and voluntary savings contributions. For the lower-ranking firm, simplification increased the likelihood that they recalled that their firm was ranked worse relative to others, reducing consumer confidence and likelihood of making a retirement savings contribution. Our work suggests that careful consideration should be made to what type of information is being emphasized when simplification is used.

From a theoretical perspective, our work contributes to the importance of informational fluency on decision-making. Consistent with previous fluency work (Landwehr and Eckmann 2020), we show that simplifying information reduces reading time and improves information comprehension and recall. However, whether simplification improved or reduced voluntary retirement contributions was highly dependent on the valence of firm comparison information. By observing both positive and negative effects of simplicity on retirement contributions in Mexico, our work provides evidence that simple interventions that improve information processing and comprehension can significantly impact behavioral outcomes in important and financially consequential domains.

At a broader theoretical and substantive level, our work provides a few insights. First, in our field experiments, we found that all nudges were similarly effective within each firm. While some performed marginally better or worse, none of the specific nudges—that is, gain frame, loss frame, wallet cutout, or concertizing the future—were statistically differentiated from the other interventions. Recent work in other domains, such as contributing to a bank savings account, getting a flu vaccine, and exercising at a gym, has found similar effects where there was no clear

evidence of one intervention significantly outperforming others (Karlan et al. 2016; Milkman et al. 2021). Taken together, our results suggest behavioral interventions within a given category, for example, simplified forms, may be relatively fungible.

Although the precise message, framing, or intervention variant may not outperform others *within* a particular firm, it is important to test similar interventions across various settings and contexts, for example, across different firms. It is natural to assume that the effects of simplification would be similar for people across all firms. In fact, our prediction surveys with policymakers, marketing experts, and general population samples from Mexico were consistent with this belief. While the beneficent effects of simplification were predicted, the potential maleficent effects were not. However, our results show the importance of testing, particularly when the goal is to scale interventions to broader populations.

Our study explores the role of simplification on voluntary retirement contributions over a 4-month period. Thus, a natural question arises: Are one-time period increases (or decreases) in voluntary contributions economically meaningful in the long run? To quantify the long-term impact of heterogeneous treatment responses on voluntary contributions across *all* firms, we begin by calculating the number of people in the various firms. First, we calculate the number of customers in the top four ranked firms, which totals to roughly 18.36 million people. Next, we calculate the number of customers in the bottom four ranked firms, which totals to about 15 million people. Given that our field experiments were conducted with a high-ranked and low-ranked firm, we could not confidently make claims about the effects of form simplification for consumers in firms that were ranked in the middle, leaving them out of our calculations.

If the form simplification intervention was scaled up to all seven firms, then using the simplified form could generate 45,096 new contributors, and \$324,360 pesos in contributions. If simplification were only applied to the top-performing firms, it would generate 66,096 new contributors, and \$477,360 in contributions. For the low-performing firms, simplification would reduce contributors by 21,000 and contributions by \$153,000. The long-term returns from an additional contribution can also be substantial. For example, for the top-performing four firms, the annual average net rate of return is 1.29%. If a 22-year-old in one of these firms were to make a one-time contribution of \$4,280.47 (the average amount contributed for a contributor in the high-ranked firm), this would compound to \$7,334.28. For the four low-ranking firms, the average annual rate of return is 0.956%. A consumer who made a one-time contribution of \$3,341.53 at age 22 would receive \$4,982.87 upon retirement. As a result, an account holder who is dissuaded from contributing because of the

statement simplification would lose a substantial amount of money in the long-term.

Of course, there are many instances when firms deliberately obfuscate negative information to prevent consumers from switching to another provider (Brown et al. 2010; Gabaix and Laibson 2006). There are also many instances where policymakers create frictions, or sludges, *unintentionally* undermining the very behavior that they intend to improve (Thaler 2018). If the goal is to improve retirement savings, then simplifying the form or ranking information on its own will not be enough for those in low-ranking firms. One potential option to prevent backfire effects could be to keep sending the more complex, status quo forms. But doing so would not sufficiently address the policy goal of *improving* retirement savings. However, voluntary contributions are not the only way to improve retirement savings.

From an information design perspective, our results highlight that policymakers and practitioners should consider improving the simplicity and transparency of multiple routes to improve retirement savings. Promisingly, our results show that when simplification is used to make it easier to see that the firm is ranked low *and* lower the perceived costs of switching to another firm (web appendix G), then simplification can be harnessed more effectively to improve retirement savings. Specifically, for low-ranking firm customers, simplification can improve the likelihood of switching to better performing alternatives.

In our context, identifying ways to encourage switching to better performing firms is an especially effective way to boost overall retirement savings. In Mexico, most retirement savings are a result of mandatory contributions (Fertig et al. 2015). Switching to a better performing firm can improve retirement savings for everyone, including those who financially constrained and cannot afford to make additional contributions to those who have outside options that surpass the rates offered by any of the retirement providers. In all cases, switching to providers offering higher return rates is beneficial for both policy and consumer welfare objectives alike. Determining how to use simplification effectively and in a way that considers individual circumstance is essential for ensuring that consumers respond in ways that are constructive and beneficial.

It is important to acknowledge that even the most successful behavioral interventions that we identify are not aimed at fully solving the lack of retirement savings. However, increasing the propensity of making regular, small savings contributions plays an important role in ensuring a more secure financial future in retirement. Account statements from retirement savings providers are a primary and inexpensive point of contact that can influence savings rates. While other interventions require regulatory changes or more expensive additional forms of communication (e.g., personalized interventions, SMS interventions, educational programs, etc.), the present intervention simply



calls for organizations to alter information design using already existing forms of communication. Any simple intervention that increases the effectiveness of account statements has important practical value. Future research should compare the effect size of the present intervention to other potential interventions. Future work may also consider whether various interventions interact and when they do so in ways that may even amplify their effectiveness, for example, simplifying the rank information and perceived switching costs, as compared to when they are used on their own.

One major question that our field experiment raises is how people choose their retirement firm in the first place. Does the importance of certain attributes, such as geographical closeness, play more of a role in the initial choice stage but becomes less important as people observe the monetary gains/losses over time? Are individuals who are more attentive or who are more financially literate less likely to pick a low-ranked firm or quicker to switch to a high-ranked firm? Gaining a better understanding of whether the firm selection process differs from the voluntary savings contribution decision process and whether the tradeoffs vary is crucial from a policy and consumer welfare standpoint.

## DATA COLLECTION STATEMENT

The main data collection for field experiments 1A and 1B was conducted in Mexico from April 2016 to March 2017, with further data collection conducted in May 2020 by representatives from ideas42, CONSAR (Mexico's government regulation body), and representatives from our two retirement providers. The data were analyzed by the first and second authors and stored in a secure server at ideas42. The data for experiment 2 were collected by the first author in March 2022 via Prolific. The data were analyzed by the first author and are on a secure server and on the Open Science framework. The data for the two field experiments contain confidential financial data from Mexico, and thus are proprietary and cannot be released publicly. We can arrange for individuals to work with the raw data for the purpose of replication on a secure computer after arranging for a non-disclosure agreement. The research team will facilitate this process for replication. The authors are happy to provide the code used for the machine learning procedures upon request. The data collection statement for all [web appendix](#) studies can be found in [web appendix J](#).

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